Sahaj Bhadja

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Summary

Software Engineer and USC MSCS graduate with 2+ years building reliable AI applications and platform services. Skilled in **C++**, Python, object-oriented design, concurrency, performance optimization, LLM systems, testing, and debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Software Engineering, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Object-Oriented Programming, Computer Networks, Machine Learning

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant AI orchestration with explicit job states, background workers, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **software reliability** across 5K+ monthly tasks.
- Investigated application and database bottlenecks for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through **performance optimization**, indexed queries, validation checks, code reviews, and production debugging.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed concurrent AI processing with model selection, OCR, provider fallbacks, and failure isolation, handling 1K+ files per batch while **automated testing** reduced instructor feedback turnaround 60% reliably.
- Built React and Flask interfaces supporting 150+ instructors and 10K+ daily requests, integrating multimodal models through REST APIs while maintaining **high-quality software** with CI/CD checks and production monitoring.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving offline reinforcement learning, improving simulated trajectory planning 18% while processing 100K+ logs through reproducible **machine-learning evaluation** pipelines and controlled baseline comparisons.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **simulation testing**, systematic failure analysis, debugging, and documented research evaluations.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered cross-platform integration modules using **object-oriented design patterns**, improving API response times 30%, reducing technical debt 20%, and strengthening maintainability across distributed Windows, Linux, and macOS deployments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds across three operating systems, supporting **debugging and code reviews**, defect resolution, and reproducible releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **real-time interfaces** while reducing staged production conflicts 33% across distributed application workflows.
- Implemented automated unit, integration, and end-to-end tests through Jenkins CI/CD and Docker staging, achieving 95%+ **build validation** coverage and reducing deployment time 50% across controlled production releases.

Skills

Languages: C++, Python, Java, TypeScript, JavaScript, SQL, C#, Bash
Systems & Performance: Data Structures, Algorithms, OOP, Design Patterns, Concurrency, Multithreading, Async Processing, Operating Systems, Linux, macOS, Memory Management, Profiling, Performance Optimization, Resource-Constrained Systems
AI & Applications: LLMs, Generative AI, Transformers, PyTorch, Multimodal AI, OCR, Reinforcement Learning, Vector Search, FAISS, REST APIs, FastAPI, Flask, React, SQLite, PostgreSQL
Quality & Delivery: Git, Code Review, Unit Testing, Integration Testing, Regression Testing, PyTest, Mockito, Debugging, Root-Cause Analysis, CI/CD, GitHub Actions, Jenkins, Docker, AWS, Azure, Agile/Scrum, Observability, Technical Documentation, AI-Assisted Development

Projects

LocalIntent Engine | *C++, Python, LLaMA-2, SQLite, Multithreading, PyTest*

- Designed a resource-aware **local intelligence runtime** that classifies user intents, invokes registered tools, preserves private context locally, and applies concurrent task scheduling, graceful fallbacks, and deterministic evaluation fixtures.

Big-Brain | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built an AI knowledge platform combining retrieval-augmented generation, semantic search, vector storage, and REST APIs to deliver **grounded responses** from domain documents through scalable cloud-hosted application workflows.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling algorithms in C++ with **interactive visualization**, validating 1K+ process combinations and reducing average turnaround time up to 25%.